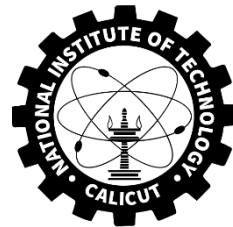


Quality, Reliability & Maintenance ME3327E



Course Instructor:

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Assistant Professor, Dept. of Mechanical Engineering

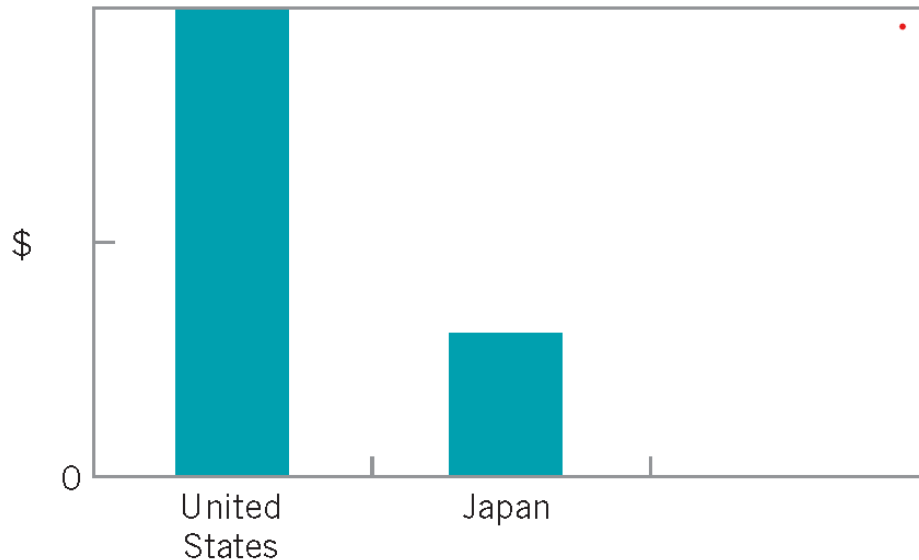
Quality

Quality means fitness for use.

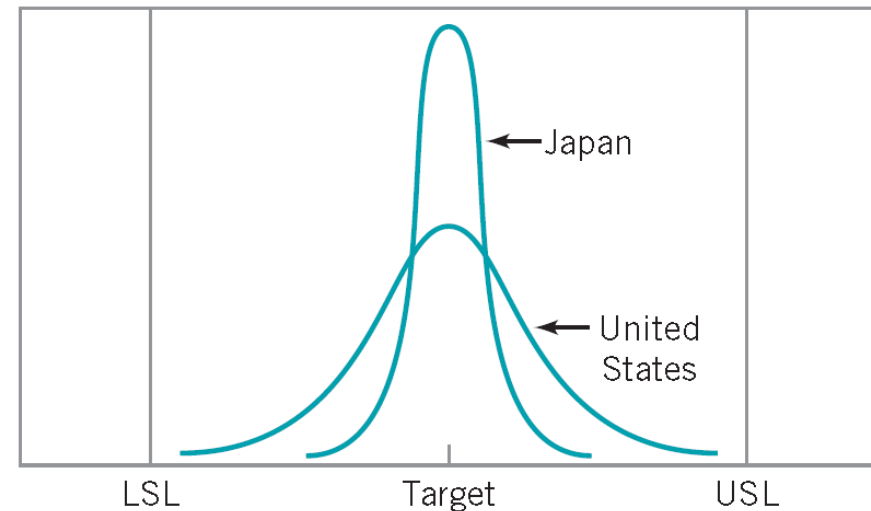
- There are two general aspects of fitness for use: quality of design and quality of conformance.

Quality is inversely proportional to variability.

- A few years ago, one of the **automobile companies** in the **United States** performed a comparative study of a transmission that was manufactured in a domestic plant and by a Japanese supplier.
- An analysis of warranty claims and repair costs indicated that there was a striking difference between the two sources of production, with the Japanese-produced transmission having much lower costs.



Warranty costs for transmissions

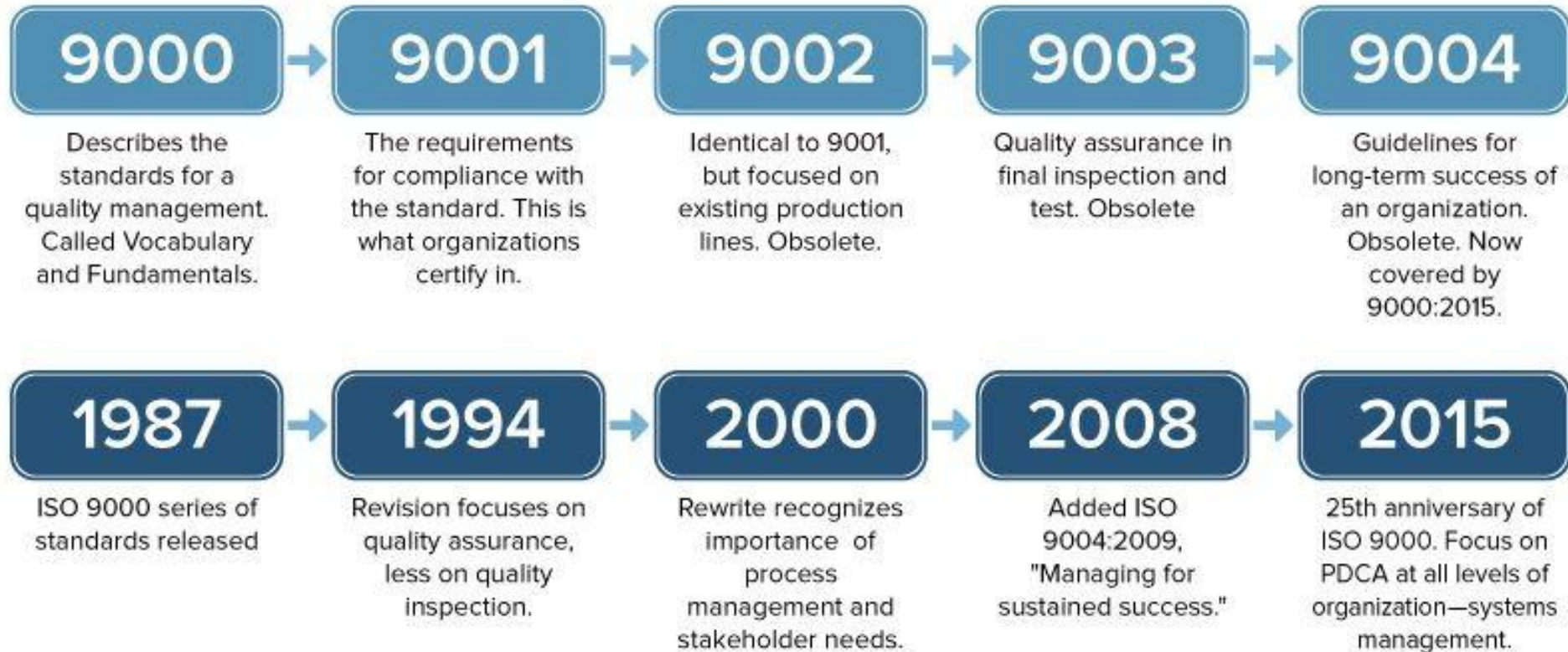


Distributions of critical dimensions for transmissions

Quality Engineering terminology

- ❑ Every product possesses a number of elements that jointly describe what the user or consumer thinks of as quality. These parameters are often called quality characteristics. Sometimes these are called critical-to-quality (CTQ) characteristics. B may be of several types:
 - Physical: length, weight, voltage, viscosity
 - Sensory: taste, appearance, color
 - Time Orientation: reliability, durability, serviceability
- ❑ Quality engineering is the set of operational, managerial, and engineering activities that a company uses to ensure that the quality characteristics of a product are at the nominal or required levels and that the variability around these desired levels is minimum.
- ❑ Other terminologies: statistical methods; variability; attributes or variables; specifications; nominal or target value; upper specification limit (USL); lower specification limit (LSL); over-the-wall; nonconforming products; nonconformity; defective; defects; concurrent engineering.

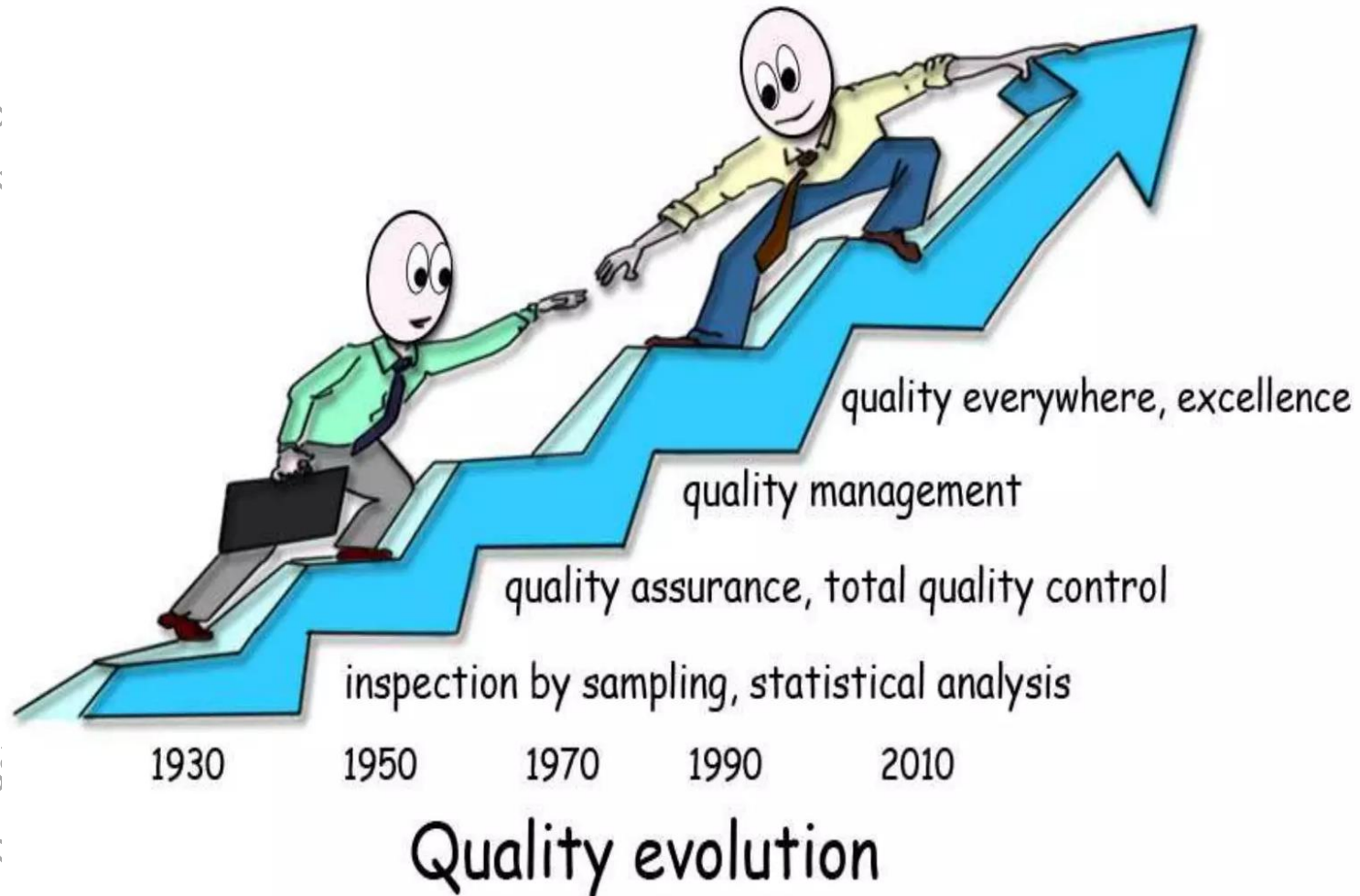
History of Quality Control and Improvement



History of Quality Control and Improvement

- ❑ **Frederick W. Taylor** introduced some principles of scientific management as mass production industries began to develop prior to 1900
- ❑ **Frank Gilbreth** and others extended this concept to the study of motion and work design
- ❑ In 1924, **Walter A. Shewhart** of the Bell Telephone Laboratories developed the statistical control chart concept, which is often considered the formal beginning of statistical quality control. Toward the end of the 1920s, **Harold F. Dodge** and **Harry G. Romig**, both of Bell Telephone Laboratories, developed statistically based acceptance sampling as an alternative to 100% inspection. By the middle of the 1930s, statistical quality-control methods were in wide use at Western Electric, the manufacturing arm of the Bell System.
- ❑ The American Society for Quality Control was formed in 1946. This organization promotes the use of quality improvement techniques for all types of products and services
- ❑ The 1950s and 1960s saw the emergence of reliability engineering
- ❑ In the 1950s, designed experiments for product and process improvement were first introduced in the United States.

History/Quality Evolution



≡ The 7 Core Pillars of QMS (ISO 9001)

ISO 9001 standard frameworks are anchored in seven fundamental Quality Management Principles (QMPs) configured for organizational cohesion:

1. **Customer Focus:** Meeting needs and striving to exceed expectations.
2. **Leadership:** Creating unity of purpose and corporate direction.
3. **Engagement of People:** Empowering competent staff at all tiers.
4. **Process Approach:** Managing activities as interrelated, logical workflows.
5. **Improvement:** Maintaining an ongoing focal commitment to optimization.
6. **Evidence-based Decisions:** Choosing paths based on systemic data analysis.
7. **Relationship Management:** Nurturing mutual value with critical partners.

Strategic Value Propositions

Implementing QMS principles is not a compliance checklist; it dictates commercial survival and high-velocity scaling:

Risk Mitigation & Resilience

Standardizing key pathways radically minimizes downstream product failure, scrap rates, regulatory litigation, and catastrophic public recalls.

Cost Reductions & Profitability

Directly optimizes internal resource allocation, removing redundant actions, reducing process variations, and conserving operational runway.

Brand Equity & Customer Trust

Delivering uniform outputs repeatedly matches expectations, creating stable organic retention rates and justifying premium pricing structures.

The Road to Operational Excellence

Align & Design:

Strategic Setup: Translate the seven QMPs into explicit, top-tier corporate objectives, establishing clear metrics of success and process benchmarks.

Process Mapping:

Practical Execution: Embed standard checkpoints inside production workflows. Establish visible, accessible document hierarchies and feedback chains.

Engage & Train:

Sustain Engagement: Decentralize standard checks. Train teams to run root-cause analysis locally, eliminating functional communication silos.

Excel & Evolve:

Peak Standards: Run data audits regularly. Execute continuous improvement loops (Kaizen, DMAIC) to transition from standard control into predictive Quality 4.0.

Dimensions of Quality



Key Components of Quality Management

Quality Planning

- Establish quality objectives and determine the necessary processes, resources, and timelines to meet them.

Quality Assurance

- Implement systematic, preventive measures and maintain documented procedures to ensure that processes consistently produce compliant outcomes.

Quality Control

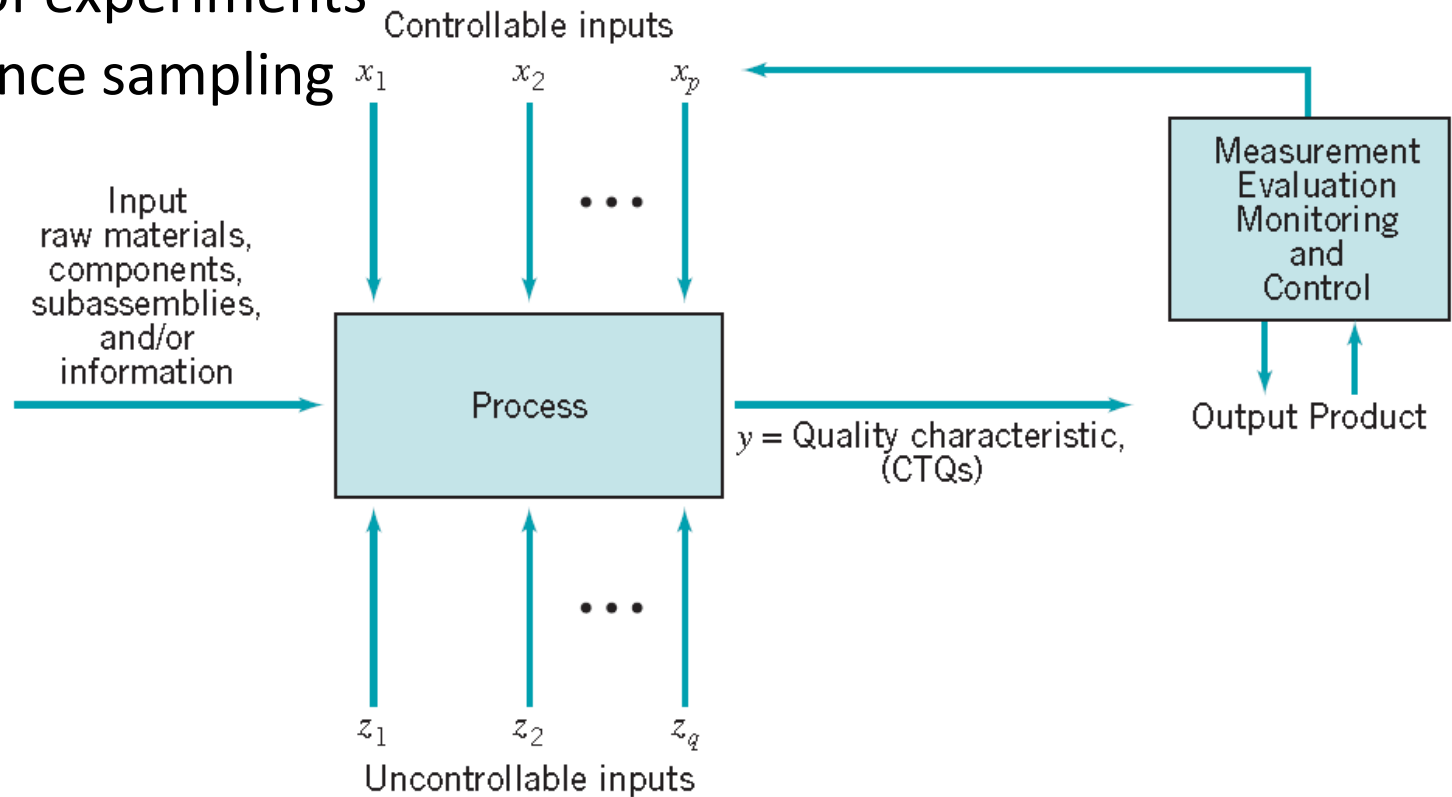
- Monitor outputs using inspections, testing, and measurement to detect deviations from specifications.

Quality Improvement

- Continuously enhance processes by identifying inefficiencies and removing root causes of defects or failures.

Statistical Methods for Quality Control and Improvement

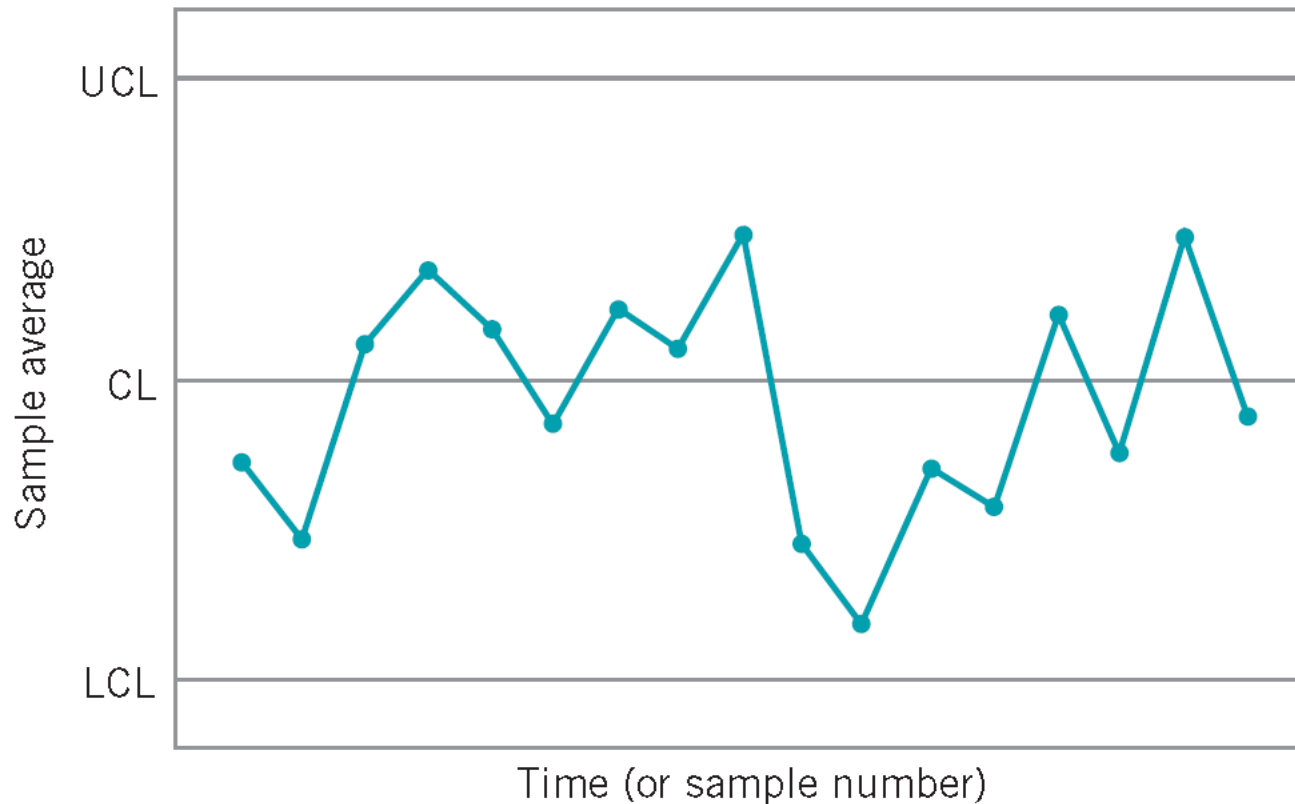
- Three major areas:
 - Statistical process control
 - Design of experiments
 - Acceptance sampling



Production process inputs and outputs

Statistical process control

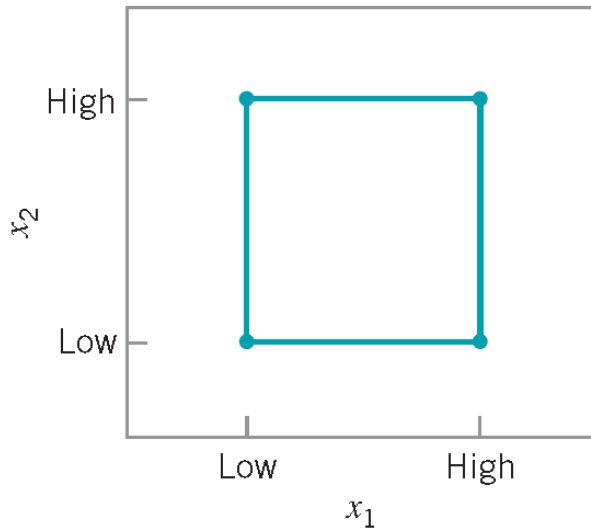
- A control chart is one of the primary techniques of statistical process control (SPC).



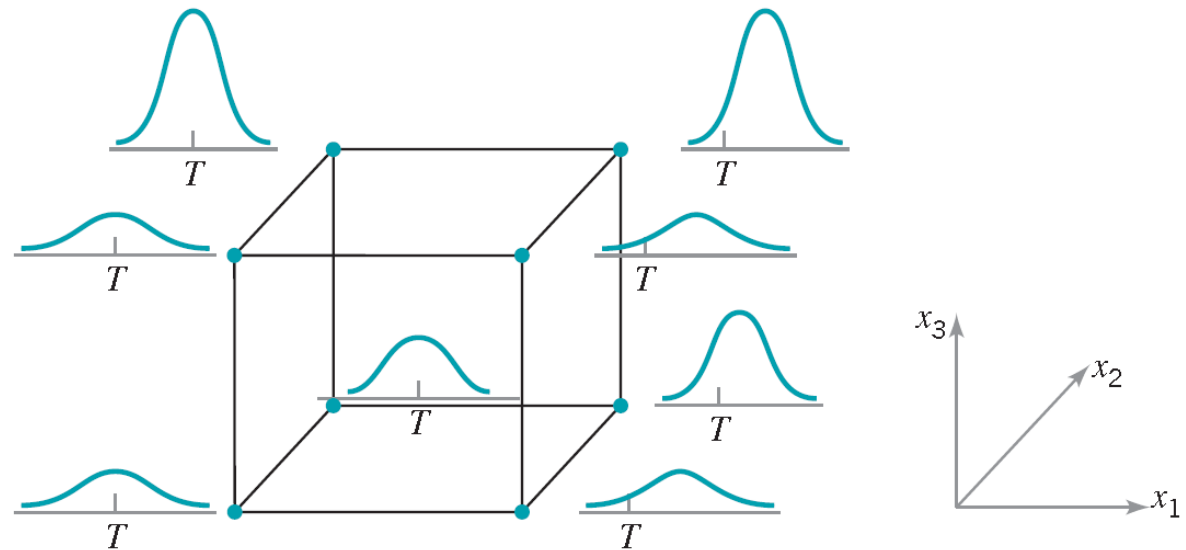
A typical control chart

Design of experiments

- Genichi Taguchi
- One major type of designed experiment is the factorial design



(a) Two factors, x_1 and x_2

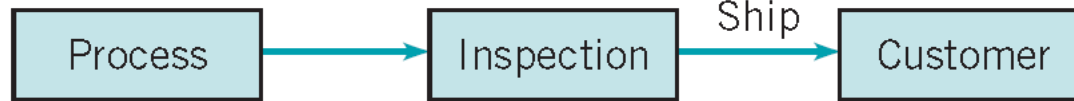


(b) Three factors, x_1 , x_2 , and x_3

Factorial designs for the process

Acceptance sampling

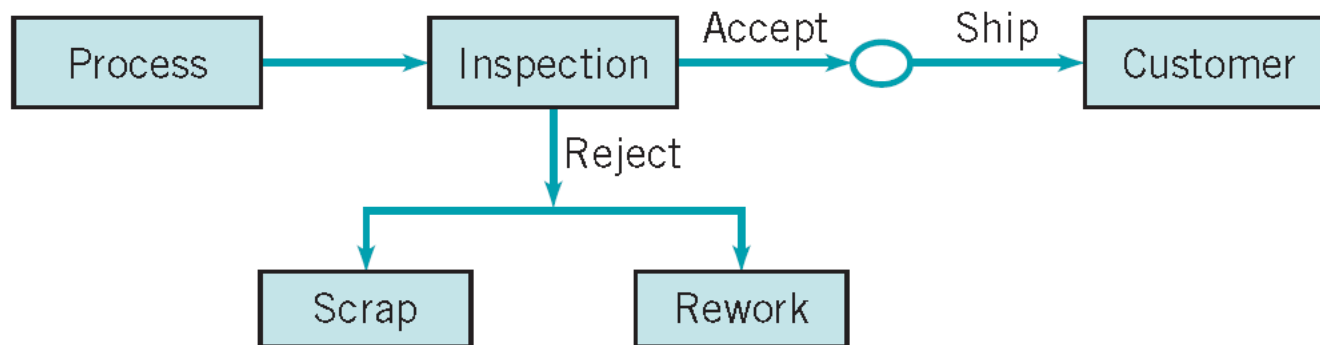
Acceptance sampling, defined as the inspection and classification of a sample of units selected at random from a larger batch or lot and the ultimate decision about disposition of the lot, usually occurs at two points: incoming raw materials or components, or final production.



(a) Outgoing inspection



(b) Receiving/incoming inspection



(c) Disposition of lots

Variations of acceptance sampling